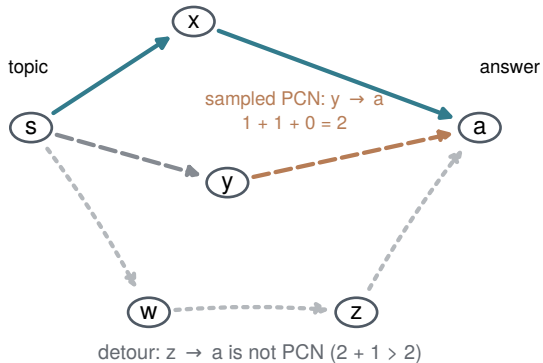


a Complete paths and actual negative sampling

selected positive path



b Supervision changes; inference is shared

Same training candidates and MLP

Original

$y = 0; w = 1$

Random Mask

equal-count random $w = 0$

PCN Mask

PCN: $y = 0; w = 0$

Positive Relabel

PCN: $y = 1; w = 1$

Training answers identify PCN only

Inference: $q + \text{topic} \rightarrow \text{candidates} \rightarrow \text{rank}$